# Syllabic frequencies reveal a strong cross-linguistic prevalence of the Sonority Sequencing Principle

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**Syllabic frequencies reveal a strong cross-linguistic prevalence of the Sonority Sequencing Principle**

Sonority is a phonological feature that describes the perceptual salience of a given phoneme relative to neighbouring ones (Blevins, 1995; S. Parker, 2008; S. G. Parker, 2002; Selkirk, 1984). While the acoustic correlates of sonority are still under debate, some physical features like intensity (dB) have been shown to strongly correlate with sonority indexes across spoken languages (Fant et al., 2000; Jany et al., 2007; S. Parker, 2017; S. G. Parker, 2002; but see Albert & Nicenboim, 2022 for a different perspective). Phonemes can be hierarchically organised according to their sonority, with those at the lower end having lower sonority and those at the higher end having higher sonority (e.g., Obstruents < Nasals < Liquids < Glides < Vowels, Clements, 1990). This hierarchy is often represented as a numerical scale (e.g., `1` for lowest sonority, `9` for highest sonority, S. Parker, 2008). According to this hierarchy, some syllables are considered well-formed across languages, whereas others are marked as ill-formed. The Sonority Sequencing Principle (SSP) is a phonotactic rule that captures how sonority constrains the ordering of phonemes within a syllable. The SSP states that each syllable should display a unique peak of sonority at the nucleus, and that sonority should rise toward the nucleus and fall away from it (S. G. Parker, 2002; Selkirk, 1984). Syllables are classified as following or violating the SSP by examining the sequence of sonority values assigned to each of its phonemes. For instance, /blif/ has a sonority contour of `2-6-8-3` (`8` being the vowel /i/, nucelus of the syllable) and is therefore considered well-formed. On the other hand, /lbif/ (`6-2-8-3`) is considered ill-formed because the onset displays a sonority dip from the first to the second phoneme, violating the progressive increase in sonority that characterizes well-formed syllables (see Table 1 for more examples).

**Table 1**

*Examples of SSP-following and SSP-violating syllables in the dataset. Sonority index contours are shown for each syllable numerically and graphically. For example, the syllable /ˈskul/ in Assamese shows a violation at onset, as sonority drops from /s/ (sonority index 3) to /k/ (sonority index 1) before rising again at nucleus /u/ (sonority index 8).*

| Language | Syllable | Original word | Sonority index | |
|---|---|---|---|---|
| SSP-following | | | | |
| Faroese | /aɻ/ | altar /ˈaɻ̩.taɹ/ | 9-6 |  |
| Finnish | /berk/ | Berg /ˈberk/ | 2-9-6-1 |  |
| Kankanaey | /bal/ | balla /balˈla/ | 2-9-6 |  |
| Moksha | /alʲ/ | альняка /ˈalʲ.nʲa.kə/ | 9-6 |  |
| Sanskrit | /beː/ | कुबेर /ku.beː.ɾɐ/ | 2-9 |  |
| SSP-violating | | | | |
| Assamese | /skul/ | স্কুল /skul/ | 3-1-8-6 |  |
| German | /amts/ | Amtsträger /ˈamtsˌtʁeːgɐ/ | 9-5-1-3 |  |
| English | /biːdz/ | beads /biːdz/ | 2-8-2-4 |  |
| Serbian-Croatian-Bosnian | /br̂ːzo/ | brzo /br̂ːzo/ | 2-6-4-9 |  |
| Mon | /hma/ | ဘှာ /hma/ | 6-5-9 |  |

Experimental evidence suggests that the SSP plays a pivotal role in processing spoken (Berent et al., 2007, 2011; Berent & Lennertz, 2010; Miozzo & Buchwald, 2013; S. Parker, 2017) and sign languages (Perlmutter, 1993; Sandler, 1993; Stone et al., 2018). Studies on newborns, infants, and adults show a consistent preference toward well-formed (SSP-following) syllables over ill-formed (SSP-violating) ones, even when

well-formed syllables are unattested in the participants' native language. For instance, English-native adults prefer /bn/ (SSP-following) as onset consonant cluster over /lf/ (SSP-violating), although both clusters are not present in English (Berent et al., 2011; Berent & Lennertz, 2010). Few-day-old newborns show different brain haemodynamic responses when listening to SSP-following or violating syllables (e.g., *bran* vs. *lbif*) suggesting that precursors of the SSP are already present at birth (García-Castro et al., 2026; Gomez et al., 2014). Considering these findings, one would expect SSP-following structures to be more frequent than SSP-violating ones across virtually all languages, but a quantitative estimate of cross-linguistic prevalence of SSP-following structures is still elusive. Violations of the SSP are not uncommon across languages (Yin et al., 2023). Examples include fricative-stop consonant clusters at syllable onset (e.g., /ʃt/, as in the German word Stein) and their reverse counterpart at coda, stop-fricative clusters (e.g., /ks/, as in the English word clicks) (Kawasaki-Fukumori et al., 1997). Languages that allow considerably more complex consonant clusters at syllable onset tend to show SSP violations more frequently. This is the case of Polish, which allows for clusters of up to three consonants at syllable onset (e.g., *chrząszcz* /ˈxʂɔw̃ʂt͡ʂ/ [beetle]) (Gussmann, 2007; Jarosz, 2017) or Georgian, in which onset consonant clusters like /md/ are attested (e.g., მწვრთნელი /mt͡sʼvrtʰneli/ [trainer]) (Crouch et al., 2023).

To our knowledge, only two studies have provided a quantitative approach toward the cross-linguistic prevalence of the SSP, both considering type frequency as an index of the prevalence of SSP-following or -violating structures. Type frequencies capture the number of times that a given structure (e.g., a certain type of consonant cluster) appears in the lexicon of a given language, or in the repertoire of allowed sequences of clusters in a given language. Using this measure, evidence of a preference for SSP-following consonant clusters has been reported across 43

languages with an overall 72% compliance with the SSP across 97 consonant clusters (Frisch, 2015). Likewise, type frequencies have been examined in a larger pool of languages (496 from 58 language families), and at least 32% of the consonant clusters in word-onset position were found to violate the SSP, compared with 30% in word-final position (Yin et al., 2023). While type frequencies have provided useful information about languages' tolerance of SSP violations (or lack thereof), they do not provide a direct estimate of how often speakers of such languages encounter SSP violations in natural speech due to two limitations. First, type frequencies do not necessarily correspond on a one-to-one basis to its frequency in running speech. Consider an imaginary language that allows for 20 consonant clusters in total, one of which is /st/. While /st/ represents a sizeable portion of the language's phonotactical repertoire (5%), how frequently speakers encounter /st/ depends on at least two factors: the number of words containing /st/, and the frequency of each of those words. If /st/ occurs in only a few words or only in low-frequency words, its actual prevalence in natural speech would be lower than suggested by the language's phonological repertoire. Second, most studies extract their data from phonological transcriptions of words extracted from lexical inventories of a given language, but such phonological transcriptions often do not provide syllabic boundaries (Yin et al., 2023). Consequently, it is not always possible to include word-medial consonant clusters in the datasets, leaving a substantial portion of syllabic structures uttered in natural speech unobserved with only word-initial and word-final syllables being analyzed.

In the present study, we aim to overcome these two pitfalls to provide a quantitative estimate of cross-linguistic SSP prevalence that more accurately reflects the frequency with which speakers encounter SSP-following and SSP-violating syllables in natural speech. To this end, we first took into account a measure of syllabic token

frequency, defined as the number of times a given syllable is uttered in a given language per million words (Cholin & Levelt, 2009; Duchon et al., 2013; Perea & Carreiras, 1998). Second, we included a large sample of syllabified phonological transcriptions for each language in our dataset, including word-initial, word-medial and word-final syllables. We then used a Bayesian modelling approach to estimate the prevalence of the SSP in this dataset which comprised 52,588 syllables from 42 languages from 8 distinct language macro-families. This approach yielded estimates of the proportion of SSP-following syllables in each language, adjusted for the relative frequency of each syllable in natural speech.

## Methods

Raw data, reproducible code, and project documentation are available in the Open Science Framework (https://osf.io/yeqms) and GitHub (https://github.com/NeuroDevCo/ssp-corpus).

### Data collection and preprocessing

Phonological transcriptions were scraped from word entries in the Wiktionary database. Wiktionary is an open-source, free database containing 10,050,751 lexical entries from 8,244 languages contributed by 4,463,976 users (as of June 10th, 2026), many of which include phonological transcriptions of words in IPA. The wikipron Python package (version 1.3.3, Lee et al., 2020) was used to scrape all available phonological transcriptions of words for all 7,859 languages included in the Glottolog database (Nordhoff & Hammarström, 2011). This resulted in approximately 1.7 million phonological transcriptions across 900 languages. Languages with less than 100 words were excluded, resulting in a final dataset of 731,145 words from 192 languages.

Available lexical frequencies for each word in each language were extracted using the *wordfreq* Python package (Speer, 2022), which provides an interface for

retrieving lexical frequency data for words in over 40 languages. When lexical frequencies were not available for a particular language (e.g., the ISO 639-3 code did not match any language available in *wordfreq*), the closest equivalent language was used, if available. Equivalent languages were identified based on their distance from the original language according to CLDR v38.11[1]. The rationale for using equivalent languages to retrieve missing lexical frequencies was that users of close languages are likely to use similar words to refer to the same objects, and lexical frequencies were therefore expected to be reasonably comparable across equivalent languages. Words with missing lexical frequency scores were excluded from the dataset.

The phonological transcriptions of words were then segmented into syllable candidates. When possible, syllable boundary markers were used, either explicit (“.”) or implicit (“ˈ” for primary stress, “ˌ” for secondary stress). In the absence of syllable markers, the word was assumed to be monosyllabic. Segments of each syllable candidate were then classified as consonants (C), vowels (V), or semi-vowels (v). Individual vowels, consecutive strings of vowels and semi-vowels (e.g., V, VV, Vv, VVv) were classified as vocalic clusters. Syllable candidates with more than one vocalic cluster (e.g., CVvCV, VCV) were flagged as multisyllabic and excluded from the dataset. This methods allowed to extract word-medial syllables when possible. Duplicated syllables within each language were removed (e.g., if the syllable /ka/ appeared in the Spanish words /ˈka.sa/ and /kaˈpo/, it was counted only once in the Spanish dataset). This procedure yielded one syllabic repertoire for each language.

[1]The CLDR system assigns a numeric value to each pair of language codes based on whether both codes are assigned to identical languages with different names (e.g., Norwegian Bokmål → Norwegian, score 0), minor regional differences (score 1-3, e.g., Norwegian Bokmål → Norwegian), and languages with significant but unproblematic regional differences (score 10-24, e.g., Afrikaans → Dutch, Wu Chinese → Mandarin Chinese), etc.

To evaluate whether each extracted syllable followed or violated the SSP, a sonority score was assigned to each phoneme on a scale from `1` to `9`, using the *panphon* Python package (version 0.22.2, Mortensen et al., 2016). Syllables with more than one peak (i.e., exhibiting a local minimum in the sonority contour) were classified as SSP-violating, and as SSP-following otherwise. Syllables with a flat sonority contour (e.g., English word sphinxs /ˈsfɪŋks/, `3-3-8-5-1-3`) were classified as SSP-following. We also run the same analysis after classifying flat sonority contours as SSP violating and found equivalent results (see Appendix B).

To calculate the frequency of each syllable, we summed the lexical frequency (counts per million words) of each word the syllable was part of in the dataset (see Duchon et al., 2013 for a similar approach). The rationale behind this approach is that each time a word is uttered, all of its syllables are uttered as well. Therefore, summing the lexical frequencies of all words containing a given syllable yields an estimate of that syllable's frequency in natural speech. The final dataset comprised 52,588 syllables from 42 languages from 8 distinct language macro-families.

**Data analysis**

We used a multilevel, Bayesian logistic regression model to estimate the probability of each syllable in the dataset following (or violating) the SSP, considering the language it belonged to, and its syllabic frequency. We modelled the response variable $y_i j$ (`1`, if the syllable was SSP-following, `0` otherwise) using a Bernoulli probabilistic distribution with probability parameter $\theta_{ij}$. We specified a linear regression model for $\theta_i$, which included fixed regression coefficients for the intercept ($\alpha$) and the slope of syllabic frequency ($\beta$), and random effects $u_j$ and $w_j$ for each fixed coefficient, respectively. We specified weakly informative priors for all parameters in the model (see Equation 1). To estimate the posterior distribution of the model we used Hamiltonian

Monte Carlo as implemented in *PyMC3* (version 6.2.0, Patil et al. (2010)), with four chains of 2,000 iterations, 1,000 of which were warm-up iterations.

**Likelihood**:

$$\text{Global prevalence of the SSP:}$$

$$y_{ij} \sim \text{Bernoulli}\ (\frac{\theta_{ij}}{1-\theta_{ij}})$$

$$\theta_{ij} = \alpha + u_j + (\beta + w_j) \cdot \text{Frequency}_{ij}$$

**Prior**:

$$\begin{aligned} \alpha, \beta &\sim \text{Normal}\ (0, 1) \\ u_j, w_j &\sim \text{Normal}\ (0, \sigma_{u,w}) \\ \sigma_{u,w} &\sim \text{HalfNormal}\ (0.5) \end{aligned} \tag{1}$$

**Where**:

$$i \text{ indexes each observation and } j \text{ indexes the language it belongs to}$$

$$y_{ij} \text{ indexes whether the syllable follows } (1) \text{ or violates } (0) \text{ the SSP}$$

$$u_j, w_j \text{ are language-specific adjustments to } \alpha \text{ and } \beta$$

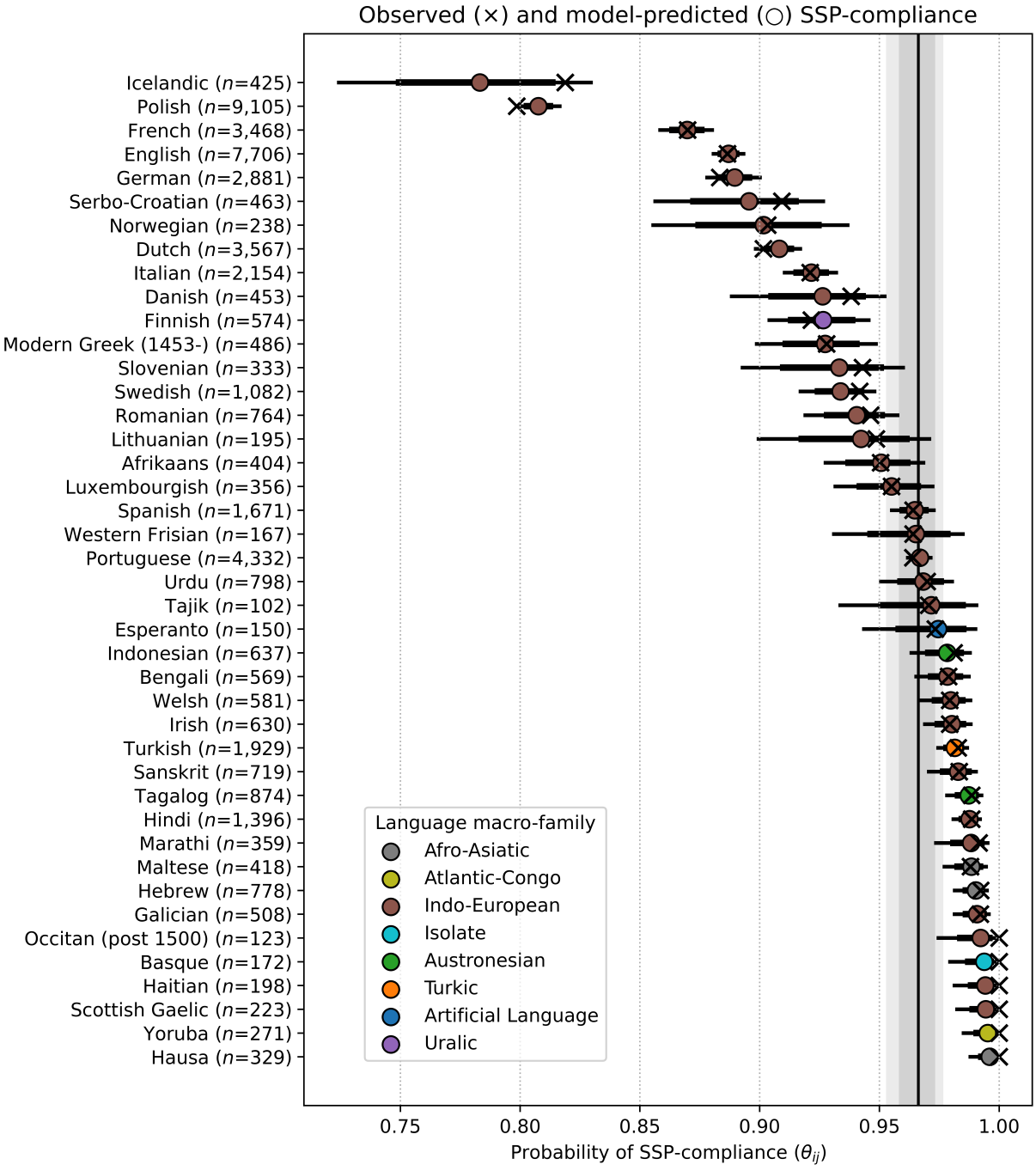
## Results

We report the median and 95% credible interval (CrI) of the posterior distribution of the model parameters of interest. The 95% CrI contains the 95% of the posterior distribution of the parameter. This interval can be interpreted as being 95% likely to contain the true value of the parameter, given the observed data and the prior distribution of the model (Kruschke & Liddell, 2018).

The global proportion of SSP-following syllables across languages was $\theta$ = 0.966 (95% CrI = [0.953, 0.977]), and the language-specific prevalence of the SSP ranged from 78.319% (Icelandic, 95% CrI = [72.353%, 81.730%]) to 99.605% (Hausa, 95% CrI = [98.720%, 99.918%]) (see Figure 1). Only 2.38% of the languages in the sample showed less than 80% of SSP prevalence, 14.29% languages showed less than 90% of SSP prevalence, and 38.10% languages showed less than 95% of SSP prevalence.

**Figure 1**

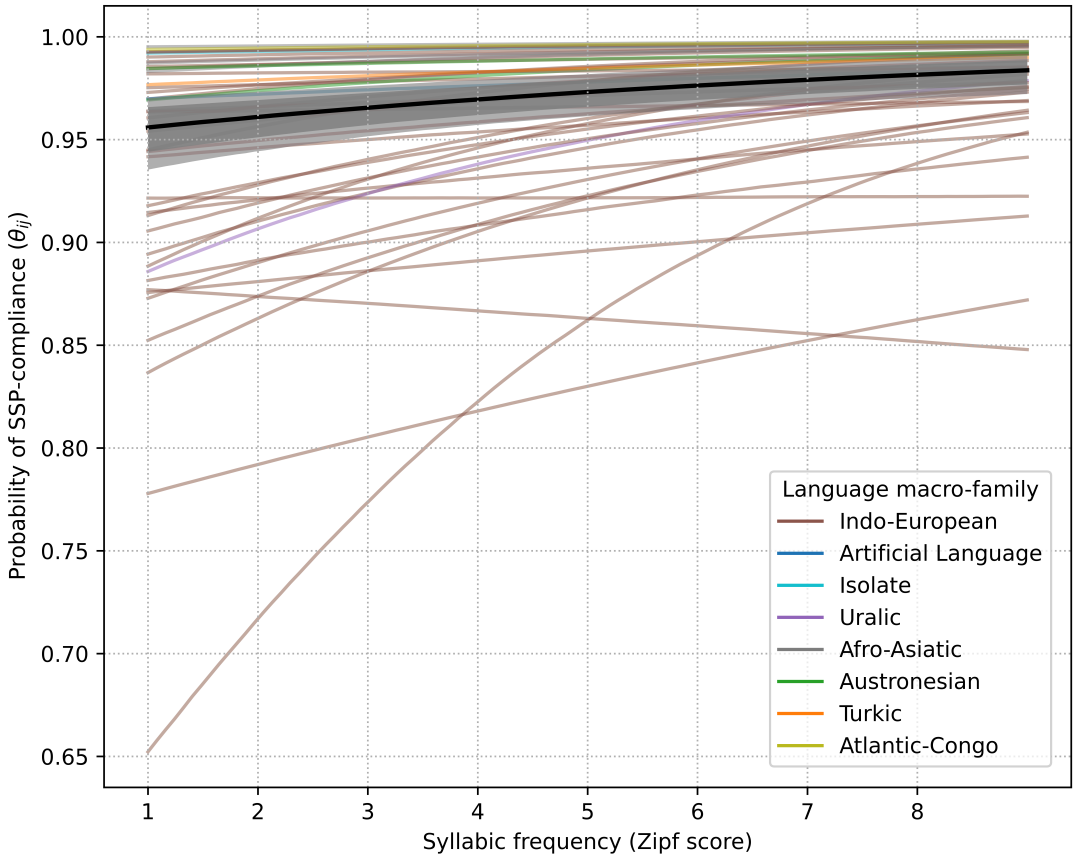
*Observed (crosses) and model-predicted (circles) estimates of SSP prevalence across languages in the dataset. Intervals indicate the 95% and 79% CrI of the posterior distribution. The number of syllables included in the dataset is indicated next to each language.*



Syllabic frequency showed a positive, non-zero association with the probability of SSP compliance ($\beta$ = 0.151, 95% CrI = [0.064, 0.246]), indicating that SSP-following syllables were, on average, more frequent than SSP-violating syllables. For every standard deviation increment in syllabic frequency (corresponding to 1.19 Zipf scores), the probability of SSP compliance increased a maximum of 0.0378 probability points (95% CrI = [0.016, 0.0614]) (see Figure 2).

**Figure 2**

*Mean predictions of SSP compliance as a function of syllabic frequency. Predicted cross-language probability of SSP compliance ($\theta_{ij}$) is showed as a black line. Thin lines indicate language-specific predictions, color-coded by language macro-family.*



## Discussion

The present study was aimed at quantifying the cross-linguistic prevalence of the Sonority Sequencing Principle (SSP). Previous literature had pointed to the SSP as a possible universal constraint to syllabic structure. This was based on previous experimental research showing that humans of all ages show a preference for SSP-following over SSP-violating structures, even if unattested in their native language (e.g., Berent & Lennertz, 2010; Berent et al., 2011), and newborns showing a similar preference in the absence of substantial experience with the phonotactical patters of their native language (Gomez et al., 2014). Such a preference could be expected to result in a lower markedness for SSP-following syllables across languages, compared to SSP-violating ones, therefore a higher frequency of occurrence for the former in natural speech. This is true to some extent. Qualitative evidence for a role of the SSP in the marking of syllabic structures had been shown in a number of languages, some of them representing edge cases of phonotatics, like Imdlawn Tashlhiyt dialect of Berber, in which syllabic nuclei may involve obstruent (voiceless) consonants (Dell & Elmedlaoui, 1985; Prince & Smolensky, 2004).

Violations of the SSP have nonetheless been reported in virtually every language (e.g., Yin et al., 2023). Because of this, the SSP cannot be taken as a hard linguistic universal, but rather as an overall preference toward certain linguistic structures that that virtually all languages display to a varying extent (Prince & Smolensky, 2004). Under this account, the posibility of the SSP being universally available as a cue for syllabification at birth becomes a quantitative question: how frequently do syllables abide by the SSP in natural speech? Quantitative estimates of SSP compliance is, to date, sparse.

In the present study, we gathered a large number of syllabified phonological transcriptions from the Wiktionary database, and derived a syllabic repertoire for each of the 42 languages in the dataset. We then evaluated the proportion of SSP-following syllables present in each syllabic repertoire, and used Bayesian modelling to generate an estimation of both cross-linguistic, and language-specific measure of SSP prevalence. Our results suggest an overwhelming presence of SSP-following syllables in natural speech across most languages in the dataset, therefore providing evidence in line with the statistical patterns of natural language reflecting an overall preference for SSP-following syllables.

[COMPARE QUANTITATIVE FINDINGS WITH PRIOR RESEARCH]

Frisch (2015) Yin et al. (2023) Al Tamimi and Al Shboul (2013)

[METHODOLOGICAL PITFALLS WE ADDRESSED]

[SYLLABIC FREQUENCIES]

Positive association with SSP compliance. Controls for the problem of selection bias? (Yin et al., 2023)

Our findings have implications for language acquisition. Syllables are a privileged linguistic unit in speech processing during the earliest stages of language acquisition (Bertoncini et al., 1988; Eimas, 1999; Jusczyk & Derrah, 1987). For instance, during as early as their first days of life humans are able to discriminate between word-forms based on the number of syllables, instead of relying on the number of phonemes or the duration of the produced utterance (Bijeljac-Babic et al., 1993). Newborns can also track the co-ocurrence of syllable-sized groups of speech sounds to chunk the continuous speech stream into word-like units (Fló et al., 2019, 2022), an ability that will later give them access to the precursors of a lexicon (Saffran et al., 1996; Saffran & Kirkham, 2018). Given the relatively limited experience newborns have accumulated with their

native language, their ability to identify syllables in speech is remarkable. Previous findings suggest that newborns and young infants are sensitive to certain constraints to syllabic structure (e.g. Bertoncini & Mehler, 1981; Gomez et al., 2014; Moon & Fifer, 1990; Santolin et al., 2024).

The present status does not argue in favour or against the universality status of the SSP, but does point to the SSP as a universally available, functionally relevant phonological feature

[LIMITATIONS]

- Possible selection bias
- Representativity sonority index, acoustic correlates
- Level of detail in phonological representations
- No direct evidence of SSP prevalence in the context of language usage (ideally, we would transcribe and syllabify a corpus and simply calculate the proportion of SSP-following syllables). Currently not possible (few datasets are syllabified).
- Althoug the present study includes a relatively large sample of languages from different language families, Indoeuropean (mostly Western) languages are overrepresented in the dataset, paralelling the same pattern observed in corpus analysis in Linguistics.
- The classification of segments into vowels and consonants ignores that in some languages, certain consonantal phonemes can act like syllabic nuclei. This is the case of post-stress /n/ in the English word “cotton” /kɒt.n̩/.

[NEXT STEPS]

[CONCLUSIONS]

## References

Al Tamimi, Y. A., & Al Shboul, Y. (2013). Is the phonotactics of the Arabic complex coda sonority-based? *Journal of King Saud University-Languages and Translation*, *25*(1), 21–33.

Albert, A., & Nicenboim, B. (2022). Modeling Sonority in Terms of Pitch Intelligibility With the Nucleus Attraction Principle. *Cognitive Science*, *46*(7), e13161. https://doi.org/10.1111/cogs.13161

Berent, I., Harder, K., & Lennertz, T. (2011). Phonological Universals in Early Childhood: Evidence from Sonority Restrictions. *Language Acquisition*, *18*(4), 281–293. https://doi.org/10.1080/10489223.2011.580676

Berent, I., & Lennertz, T. (2010). Universal constraints on the sound structure of language: Phonological or acoustic? *Journal of Experimental Psychology: Human Perception and Performance*, *36*(1), 212–223. https://doi.org/10.1037/a0017638

Berent, I., Steriade, D., Lennertz, T., & Vaknin, V. (2007). What we know about what we have never heard: Evidence from perceptual illusions. *Cognition*, *104*(3), 591–630. https://doi.org/10.1016/j.cognition.2006.05.015

Bertoncini, J., Bijeljac-Babic, R., Jusczyk, P. W., Kennedy, L. J., & Mehler, J. (1988). An investigation of young infants' perceptual representations of speech sounds. *Journal of Experimental Psychology: General*, *117*(1), 21.

Bertoncini, J., & Mehler, J. (1981). Syllables as units in infant speech perception. *Infant Behavior and Development*, *4*, 247–260.

Bijeljac-Babic, R., Bertoncini, J., & Mehler, J. (1993). How do 4-day-old infants categorize multisyllabic utterances? *Developmental Psychology*, *29*(4), 711.

Blevins, J. (1995). The Syllable in Phonological Theory. In J. Goldsmith (Ed.), *Handbook of phonological theory* (Vol. 1, pp. 206–244). Blackwell.

Cholin, J., & Levelt, W. J. (2009). Effects of syllable preparation and syllable frequency in speech production: Further evidence for syllabic units at a post-lexical level. *Language and Cognitive Processes*, *24*(5), 662–684.

Clements, G. N. (1990). The role of the sonority cycle in core syllabification. *Papers in Laboratory Phonology*, 283–333. https://doi.org/10.1017/cbo9780511627736.017

Crouch, C., Katsika, A., & Chitoran, I. (2023). Sonority sequencing and its relationship to articulatory timing in Georgian. *Journal of the International Phonetic Association*, *53*(3), 1049–1072.

Dell, F., & Elmedlaoui, M. (1985). Syllabic consonants and syllabification in imdlawn tashlhiyt berber. *Journal ofAfrican Languages and Linguistics*, *7*, 105–130.

Duchon, A., Perea, M., Sebastián-Gallés, N., Martí, A., & Carreiras, M. (2013). EsPal: One-stop shopping for spanish word properties. *Behavior Research Methods*, *45*(4), 1246–1258.

Eimas, P. D. (1999). Segmental and syllabic representations in the perception of speech by young infants. *The Journal of the Acoustical Society of America*, *105*(3), 1901–1911.

Fant, G., Kruckenberg, A., & Liljencrants, J. (2000). Acoustic-phonetic analysis of prominence in swedish. In *Intonation: Analysis, modelling and technology* (pp. 55–86). Springer.

Fló, A., Benjamin, L., Palu, M., & Dehaene-Lambertz, G. (2022). Sleeping neonates track transitional probabilities in speech but only retain the first syllable of words. *Scientific Reports*, *12*(1), 4391.

Fló, A., Brusini, P., Macagno, F., Nespor, M., Mehler, J., & Ferry, A. L. (2019). Newborns are sensitive to multiple cues for word segmentation in continuous speech. *Developmental Science*, *22*(4), e12802.

Frisch, S. A. (2015). A preliminary investigation of quantitative patterns in sonority sequencing. *Italian Journal of Linguistics*, *27*(1), 9–27.

García-Castro, G., Chartier, F., Sofocleous, C., Piracún-García, L., Gervain, Judit, Alarcón Allen, A., & Santolin, C. (2026). *Sonority constraints in non-linguistic sounds*.

Gomez, D., Berent, I., Benavides-Varela, S., Bion, R., Cattarossi, L., Nespor, M., & Mehler, J. (2014). Language universals at birth. *Proceedings of the National Academy of Sciences*, *111*(16), 5837–5841.

Gussmann, E. (2007). *The phonology of polish*. OUP Oxford.

Jany, C., Gordon, M., Nash, C. M., & Takara, N. (2007). How universal is the sonority hierarchy?: A cross-linguistic acoustic study. *Proceedings of the ICPhS*, 1401–1404.

Jarosz, G. (2017). Defying the stimulus: Acquisition of complex onsets in polish. *Phonology*, *34*(2), 269–298.

Jusczyk, P. W., & Derrah, C. (1987). Representation of speech sounds by young infants. *Developmental Psychology*, *23*(5), 648.

Kawasaki-Fukumori, H., Ohala, J., Eliasson, S., & Jahr, E. (1997). Alternatives to the sonority hierarchy for explaining segmental sequential constraints. *Language and Its Ecology: Essays in Memory of Einar Haugen*, *100*, 343.

Kruschke, J. K., & Liddell, T. M. (2018). The Bayesian New Statistics: Hypothesis testing, estimation, meta-analysis, and power analysis from a Bayesian perspective. *Psychonomic Bulletin & Review*, *25*(1), 178–206.

Lee, J. L., Ashby, L. F., Garza, M. E., Lee-Sikka, Y., Miller, S., Wong, A., McCarthy, A. D., & Gorman, K. (2020). Massively multilingual pronunciation modeling with WikiPron. *Proceedings of the Twelfth Language Resources and Evaluation Conference*, 4223–4228.

Miozzo, M., & Buchwald, A. (2013). On the nature of sonority in spoken word production: Evidence from neuropsychology. *Cognition*, *128*(3), 287–301.

Moon, C., & Fifer, W. P. (1990). Syllables as signals for 2-day-old infants. *Infant Behavior and Development*, *13*(3), 377–390.

Mortensen, D. R., Littell, P., Bharadwaj, A., Goyal, K., Dyer, C., & Levin, L. (2016). Panphon: A resource for mapping IPA segments to articulatory feature vectors. *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, 3475–3484.

Nordhoff, S., & Hammarström, H. (2011). Glottolog/Langdoc: Defining dialects, languages, and language families as collections of resources. In T. Kauppinen, L. C. Pouchard, & K. Keßler (Eds.), *Proceedings of the first international workshop on linked science 2011 (LISC2011), bonn, germany, october 24, 2011* (pp. 1–7). CEUR Workshop Proceedings.

Parker, S. (2008). Sound level protrusions as physical correlates of sonority. *Journal of Phonetics*, *36*(1), 55–90. https://doi.org/10.1016/j.wocn.2007.09.003

Parker, S. (2017). Sounding out Sonority. *Language and Linguistics Compass*, *11*(9), e12248. https://doi.org/10.1111/lnc3.12248

Parker, S. G. (2002). *Quantifying the sonority hierarchy*. University of Massachusetts Amherst.

Patil, A., Huard, D., & Fonnesbeck, C. J. (2010). PyMC: Bayesian stochastic modelling in python. *Journal of Statistical Software*, *35*, 1–81.

Perea, M., & Carreiras, M. (1998). Effects of syllable frequency and syllable neighborhood frequency in visual word recognition. *Journal of Experimental Psychology: Human Perception and Performance*, *24*(1), 134.

Perlmutter, D. M. (1993). Sonority and syllable structure in american sign language. In *Current issues in ASL phonology* (pp. 227–261). Elsevier.

Prince, A., & Smolensky, P. (2004). Optimality Theory: Constraint interaction in generative grammar. *Optimality Theory in Phonology: A Reader*, 1–71. https://doi.org/10.1002/9780470756171.ch1

Saffran, J. R., Aslin, R. N., & Newport, E. L. (1996). Statistical learning by 8-month-old infants. *Science*, *274*(5294), 1926–1928.

Saffran, J. R., & Kirkham, N. Z. (2018). Infant statistical learning. *Annual Review of Psychology*, *69*, 181–203.

Sandler, W. (1993). A sonority cycle in american sign language. *Phonology*, *10*(2), 243–279.

Santolin, C., Zacharaki, K., Toro, J. M., & Sebastian-Galles, N. (2024). Abstract processing of syllabic structures in early infancy. *Cognition*, *244*, 105663.

Selkirk, E. (1984). On the major class features and syllable theory. *Language Sound Structure*.

Speer, R. (2022). *Rspeer/wordfreq: V3.0*. Zenodo. https://doi.org/10.5281/zenodo.7199437

Stone, A., Petitto, L.-A., & Bosworth, R. (2018). Visual sonority modulates infants' attraction to sign language. *Language Learning and Development*, *14*(2), 130–148.

Yin, R., van de Weijer, J., & Round, E. R. (2023). Frequent violation of the sonority sequencing principle in hundreds of languages: How often and by which sequences? *Linguistic Typology*, *27*(2), 381–403.

## Appendix A

## Number of SSP-following and SSP-violating syllables per language

**Table A1**

*Number of SSP-following and SSP-violating syllables for each language in the dataset.*

| ISO-639-3 | Words | Syllables | SSP+ | % | ISO-639-3 | Words | Syllables | SSP+ | % |
|---|---|---|---|---|---|---|---|---|---|
| eus | 10,555 | 172 | 172 | 100.0% | spa | 106,961 | 1,671 | 1,611 | 96.4% |
| hau | 738 | 329 | 329 | 100.0% | fry | 425 | 167 | 161 | 96.4% |
| oci | 243 | 123 | 123 | 100.0% | por | 99,468 | 4,332 | 4,176 | 96.4% |
| gla | 1,112 | 223 | 223 | 100.0% | ltz | 860 | 356 | 340 | 95.5% |
| hat | 848 | 198 | 198 | 100.0% | afr | 619 | 404 | 384 | 95.0% |
| yor | 1,311 | 271 | 271 | 100.0% | lit | 279 | 195 | 185 | 94.9% |
| heb | 948 | 778 | 772 | 99.2% | ron | 2,126 | 764 | 723 | 94.6% |
| glg | 3,477 | 508 | 504 | 99.2% | slv | 1,704 | 333 | 314 | 94.3% |
| mar | 703 | 359 | 356 | 99.2% | swe | 1,624 | 1,082 | 1,019 | 94.2% |
| tgl | 13,663 | 874 | 864 | 98.9% | dan | 1,399 | 453 | 425 | 93.8% |
| hin | 5,072 | 1,396 | 1,380 | 98.9% | ell | 3,104 | 486 | 451 | 92.8% |
| mlt | 3,777 | 418 | 413 | 98.8% | fin | 73,343 | 574 | 529 | 92.2% |
| san | 2,831 | 719 | 707 | 98.3% | ita | 15,211 | 2,154 | 1,984 | 92.1% |
| tur | 5,108 | 1,929 | 1,896 | 98.3% | hbs | 8,822 | 463 | 421 | 90.9% |
| ind | 3,851 | 637 | 625 | 98.1% | nor | 451 | 238 | 215 | 90.3% |
| gle | 4,682 | 630 | 617 | 97.9% | nld | 11,340 | 3,567 | 3,216 | 90.2% |
| cym | 5,150 | 581 | 569 | 97.9% | eng | 36,929 | 7,706 | 6,832 | 88.7% |
| ben | 2,771 | 569 | 557 | 97.9% | deu | 13,142 | 2,881 | 2,545 | 88.3% |
| epo | 5,975 | 150 | 146 | 97.3% | fra | 17,659 | 3,468 | 3,017 | 87.0% |
| tgk | 258 | 102 | 99 | 97.1% | isl | 2,346 | 425 | 348 | 81.9% |
| urd | 1,088 | 798 | 774 | 97.0% | pol | 123,564 | 9,105 | 7,271 | 79.9% |

## Appendix B

## Results after classifying flat sonority contours as SSP-violating

**Figure B1**

*Observed (crosses) and model-predicted (circles) estimates of SSP prevalence across languages in the dataset. Intervals indicate the 95% and 79% CrI of the posterior distribution. The number of syllables included in the dataset is indicated next to each language.*

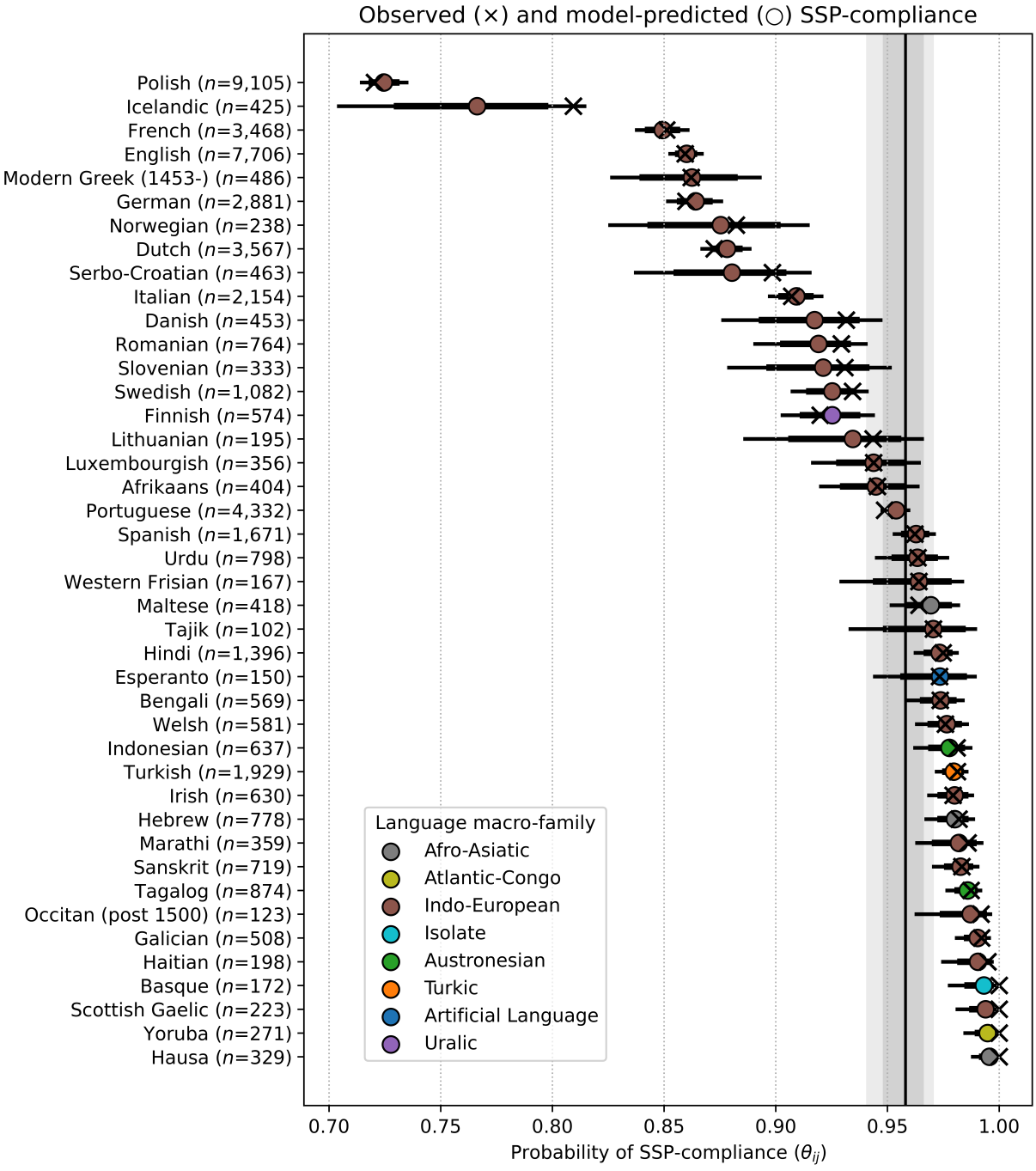


**Figure B2**

*Mean predictions of SSP compliance as a function of syllabic frequency. Predicted cross-language probability of SSP compliance ($\theta_i j$) is showed as a black line. Thin lines indicate language-specific predictions, color-coded by language macro-family.*

